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Judul: Boogle Slang Dictionary menggunakan Trie

Import Library

Code:

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
#include <string.h>
```

```
#include <ctype.h>
```

Penjelasan:

stdio.h: digunakan untuk input-output.

stdlib.h: untuk alokasi memori dinamis (malloc, free).

string.h: untuk operasi string seperti strlen(), strcpy().

ctype.h: untuk manipulasi karakter seperti tolower().

Konstanta

Code:

```
#define MAX_CHAR 26
```

```
#define MAX_LEN 105
```

Penjelasan:

MAX_CHAR: jumlah karakter alfabet (a-z).

MAX_LEN: maksimal panjang kata atau kalimat.

Struktur Trienode

Code:

```
typedef struct TrieNode {  
    struct TrieNode* children[MAX_CHAR];  
    char* description;  
    int isEndOfWord;  
} TrieNode;
```

Penjelasan:

children[MAX_CHAR]: array node anak sebanyak 26, masing-masing untuk setiap huruf a-z.

description: menyimpan deskripsi slang word.

isEndOfWord: penanda apakah sebuah node merupakan akhir dari kata.

Membuat node baru

Code:

```
TrieNode* createNode() {  
    TrieNode* newNode = (TrieNode*)malloc(sizeof(TrieNode));  
    newNode->isEndOfWord = 0;  
    newNode->description = NULL;  
    for (int i = 0; i < MAX_CHAR; i++) newNode->children[i] = NULL;  
    return newNode;  
}
```

Penjelasan:

Membuat node Trie baru dengan semua pointer anak diset NULL, deskripsi NULL, isEndOfWord = 0.

Validasi input word

Code:

```
int isValidWord(char* word) {  
    if (strlen(word) <= 1 || strchr(word, ' ')) return 0;  
    return 1;  
}
```

Penjelasan:

Mengecek apakah panjang kata lebih dari 1 karakter dan tidak mengandung spasi.

Menghitung jumlah kata

Code:

```
int wordCount(char* sentence) {  
    int count = 0, inWord = 0;  
    for (int i = 0; sentence[i]; i++) {  
        if (!isspace(sentence[i]) && !inWord) {  
            inWord = 1;  
            count++;  
        } else if (isspace(sentence[i])) {  
            inWord = 0;  
        }  
    }  
}
```

```

    }

    return count;
}

```

Penjelasan:

Menghitung jumlah kata dalam deskripsi, digunakan untuk validasi deskripsi minimal 2 kata.

Insert kata

Code:

```

void insert(TrieNode* root, char* word, char* desc) {

    TrieNode* curr = root;

    for (int i = 0; word[i]; i++) {

        int idx = tolower(word[i]) - 'a';

        if (!curr->children[idx])

            curr->children[idx] = createNode();

        curr = curr->children[idx];

    }

    curr->isEndOfWord = 1;

    if (curr->description)

        free(curr->description);

    curr->description = (char*)malloc(strlen(desc) + 1);

    strcpy(curr->description, desc);

}

```

Penjelasan:

Memasukkan kata baru ke Trie huruf per huruf, Jika sudah ada, update deskripsi lama. Alokasi memori baru untuk deskripsi.

Mencari kata di trie

Code:

```
TrieNode* search(TrieNode* root, char* word) {  
  
    TrieNode* curr = root;  
  
    for (int i = 0; word[i]; i++) {  
  
        int idx = tolower(word[i]) - 'a';  
  
        if (!curr->children[idx]) return NULL;  
  
        curr = curr->children[idx];  
  
    }  
  
    return (curr && curr->isEndOfWord) ? curr : NULL;  
}
```

Penjelasan:

Melakukan pencarian kata di Trie dan mengembalikan node jika ketemu, atau NULL jika tidak.

Menampilkan semua kata di trie

Code:

```
void printAll(TrieNode* root, char* prefix, int level, int* count) {  
  
    if (root->isEndOfWord) {  
  
        prefix[level] = '\0';  
  
        printf("%d. %s\n", ++(*count), prefix);  
  
    }  
  
    for (int i = 0; i < 26; i++) {  
  
        if (root->children[i])  
            printAll(root->children[i], prefix, level + 1, count);  
  
    }  
}
```

```

    }

    for (int i = 0; i < MAX_CHAR; i++) {
        if (root->children[i]) {
            prefix[level] = 'a' + i;
            printAll(root->children[i], prefix, level + 1, count);
        }
    }
}

```

Penjelasan:

Traversal rekursif menampilkan semua slang word secara berurut.

Menampilkan kata dengan prefix

Code:

```

void printWithPrefix(TrieNode* root, char* prefix) {
    TrieNode* curr = root;

    for (int i = 0; prefix[i]; i++) {
        int idx = tolower(prefix[i]) - 'a';
        if (!curr->children[idx]) {
            printf("There is no prefix \"%s\" in the dictionary.\n", prefix);
            return;
        }
        curr = curr->children[idx];
    }
}

```

```

    }

    int count = 0;

    char buffer[MAX_LEN];

    strcpy(buffer, prefix);

    printAll(curr, buffer, strlen(prefix), &count);

    if (count == 0)

        printf("There is no prefix \"%s\" in the dictionary.\n", prefix);
}

```

Penjelasan:

Traversal mencari node prefix terlebih dahulu dan Memanggil printAll() mulai dari node prefix.

Menu 1

Code:

```

void releaseSlang(TrieNode* root) {

    char word[MAX_LEN], desc[MAX_LEN];

    do {

        printf("Input a new slang word [Must be more than 1 characters and contains no space]: ");

        scanf("%s", word);

    } while (!isValidWord(word));

    do {

        printf("Input a new slang word description [Must be more than 2 words]: ");

```

```

        scanf("%[^\n]", desc);
    } while (wordCount(desc) < 2);

    TrieNode* node = search(root, word);

    insert(root, word, desc);

    if (node)

        printf("\nSuccessfully updated a slang word.\n");

    printf("Press enter to continue...");

    getchar(); getchar();
}

```

Penjelasan:

Validasi input slang word, validasi input deskripsi, dan jika ada kata yang sudah ada maka update deskripsi

Menu 2

Code:

```

void searchSlang(TrieNode* root) {

    char word[MAX_LEN];

    do {

        printf("Input a slang word to be searched [Must be more than 1 characters and contains no space]: ");

        scanf("%[^\n]", word);

    } while (!isValidWord(word));
}

```



```

TrieNode* node = search(root, word);

if (node) {

    printf("\nSlang word : %s\n", word);

    printf("Description : %s\n", node->description);

} else {

    printf("\nThere is no word \"%s\" in the dictionary.", word);

}

printf("\nPress enter to continue...");

getchar(); getchar();

}

```

Penjelasan:

Saat user memasukkan slang word, jika ditemukan maka tampilkan deskripsi dan jika tidak ditemukan tampilkan pesan kosong / tidak ada.

Menu 3

Code:

```

void viewPrefixWords(TrieNode* root) {

    char prefix[MAX_LEN];

    printf("Input a prefix to be searched: ");

    scanf("%[^\\n]", prefix);

    printf("\n");

    printWithPrefix(root, prefix);

```

```
    printf("Press enter to continue...");  
  
    getchar(); getchar();  
  
}
```

Penjelasan:

Saat user memasukkan prefix, tampilkan semua kata yang dimulai dengan prefix

Menu 4

Code:

```
void viewAllWords(TrieNode* root) {  
  
    int count = 0;  
  
    char buffer[MAX_LEN];  
  
    printAll(root, buffer, 0, &count);  
  
    if (count == 0) {  
  
        printf("There is no slang word yet in the dictionary.");  
  
    }else{  
  
        printf("List of all slang words in the dictionary:\n");  
  
    }  
  
    printf("\nPress enter to continue...");  
  
    getchar(); getchar();  
  
}
```

Penjelasan:

Tampilkan semua slang word yang tersimpan di Trie.

Menu utama

Code:

```
void menu() {  
  
    printf("Boogle Slang Dictionary\n");  
  
    printf("=====\n");  
  
    printf("1. Release a new slang word\n");  
  
    printf("2. Search a slang word\n");  
  
    printf("3. View all slang words starting with a certain prefix word\n");  
  
    printf("4. View all slang words\n");  
  
    printf("5. Exit\n");  
  
    printf("Choose >> ");  
  
}
```

Penjelasan:

Menampilkan tampilan pilihan menu utama.

Fungsi dari main

Code:

```
int main() {  
  
    TrieNode* root = createNode();  
  
    int choice;  
  
    do {  
  
        system("clear || cls");
```

```
menu();

scanf("%d", &choice);

printf("\n");

switch (choice) {

    case 1:

        releaseSlang(root);

        break;

    case 2:

        searchSlang(root);

        break;

    case 3:

        viewPrefixWords(root);

        break;

    case 4:

        viewAllWords(root);

        break;

    case 5:

        printf("Thank you... Have a nice day :)\n");

        break;

    default:

        printf("Invalid option.\n");

        break;

}

} while (choice != 5);
```

```
    return 0;

}
```

Penjelasan:

Looping program sampai user memilih Exit (pilihan 5) dan Memanggil fungsi-fungsi sesuai menu.

Test Menu 1

```
Boogle Slang Dictionary
=====
1. Release a new slang word
2. Search a slang word
3. View all slang words starting with a certain prefix word
4. View all slang words
5. Exit
Choose >> 1

Input a new slang word [Must be more than 1 characters and contains no space]: daijobu
Input a new slang word description [Must be more than 2 words]: it means everything is okay
Press enter to continue...
```

Test Menu 2

```
Boogle Slang Dictionary
=====
1. Release a new slang word
2. Search a slang word
3. View all slang words starting with a certain prefix word
4. View all slang words
5. Exit
Choose >> 2

Input a slang word to be searched [Must be more than 1 characters and contains no space]: rizz

Slang word   : rizz
Description  : charisma or charm

Press enter to continue...|
```

```
Boogle Slang Dictionary
=====
1. Release a new slang word
2. Search a slang word
3. View all slang words starting with a certain prefix word
4. View all slang words
5. Exit
Choose >> 2

Input a slang word to be searched [Must be more than 1 characters and contains no space]: dazz

Slang word   : dazz
Description  : to amaaze or stun

Press enter to continue...|
```

```
Boogle Slang Dictionary
=====
1. Release a new slang word
2. Search a slang word
3. View all slang words starting with a certain prefix word
4. View all slang words
5. Exit
Choose >> 2

Input a slang word to be searched [Must be more than 1 characters and contains no space]: bombardirocrodilo

Slang word : bombardirocrodilo
Description : its mean a flying crocodile

Press enter to continue...|
```

```
Boogle Slang Dictionary
=====
1. Release a new slang word
2. Search a slang word
3. View all slang words starting with a certain prefix word
4. View all slang words
5. Exit
Choose >> 2

Input a slang word to be searched [Must be more than 1 characters and contains no space]: skibiditoilet

Slang word : skibiditoilet
Description : its a brainroot

Press enter to continue...|
```

```
Boogle Slang Dictionary
=====
1. Release a new slang word
2. Search a slang word
3. View all slang words starting with a certain prefix word
4. View all slang words
5. Exit
Choose >> 2

Input a slang word to be searched [Must be more than 1 characters and contains no space]: tungtungungsahur

Slang word : tungtungungsahur
Description : its a wood with a face

Press enter to continue...|
```

Test Menu 3

```
Boogle Slang Dictionary
=====
1. Release a new slang word
2. Search a slang word
3. View all slang words starting with a certain prefix word
4. View all slang words
5. Exit
Choose >> 3

Input a prefix to be searched: y

1. yeet
Press enter to continue...|
```

```
Boogle Slang Dictionary
=====
1. Release a new slang word
2. Search a slang word
3. View all slang words starting with a certain prefix word
4. View all slang words
5. Exit
Choose >> 3

Input a prefix to be searched: da

1. daijobu
2. dazz
Press enter to continue...|
```

```
Boogle Slang Dictionary
=====
1. Release a new slang word
2. Search a slang word
3. View all slang words starting with a certain prefix word
4. View all slang words
5. Exit
Choose >> 3

Input a prefix to be searched: s

1. simp
2. skibiditoilet
3. sus
Press enter to continue...|
```

```
Boogle Slang Dictionary
=====
1. Release a new slang word
2. Search a slang word
3. View all slang words starting with a certain prefix word
4. View all slang words
5. Exit
Choose >> 3

Input a prefix to be searched: cri

1. cringe
Press enter to continue...|
```

```
Boogle Slang Dictionary
=====
1. Release a new slang word
2. Search a slang word
3. View all slang words starting with a certain prefix word
4. View all slang words
5. Exit
Choose >> 3

Input a prefix to be searched: ca

1. cap
Press enter to continue...|
```

Test Menu 4

```
Boogle Slang Dictionary
=====
1. Release a new slang word
2. Search a slang word
3. View all slang words starting with a certain prefix word
4. View all slang words
5. Exit
Choose >> 4

1. bombardirocrodilo
2. cap
3. cringe
4. daijobu
5. daz
6. drip
7. goat
8. lit
9. pog
10. rizz
11. simp
12. skibiditoilet
13. sus
14. tungtungungahur
15. yeet
List of all slang words in the dictionary:

Press enter to continue...|
```